



**INTRODUCTORY PHYSICS LABORATORY
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EXPERIMENT: PROJECTILE MOTION

Objectives:

1. To understand the concept of projectile motion.
2. To determine the initial velocity (v_0) dan and maximum height (h_{maks}).
3. To understand the effect of the elevation angle (θ_0) on the maximum height (h_{maks}) and maximum horizontal distance (x_{maks}).

EXPERIMENT SUMMARY

INTRODUCTION

Projectile motion is one of the most common examples of two dimensional motion encountered in everyday life, such as the trajectory of a kicked ball, the stream of water from a hose, or the launch of a projectile. According to Halliday, Resnick, and Walker [1], projectile motion refers to the motion of an object that is projected with an initial velocity at a certain angle relative to the horizontal and subsequently moves only under the influence of gravity when air resistance is neglected. Due to the effect of gravity, the object's path forms a parabolic curve, therefore, projectile motion is often referred to as parabolic motion.

The path traveled by an object during its motion is called its trajectory. Under real conditions, the trajectory is affected by air resistance, causing the object's path to deviate from an ideal parabolic shape. However, to simplify the analysis in this experiment, air resistance is neglected, and the projectile is assumed to be influenced only by its weight. Under this assumption, the projectile motion can be analyzed using Newton's Second Law. Since no force acts in the horizontal direction, the acceleration along the x-axis is zero, while the acceleration in the vertical direction is equal to the acceleration due to gravity.

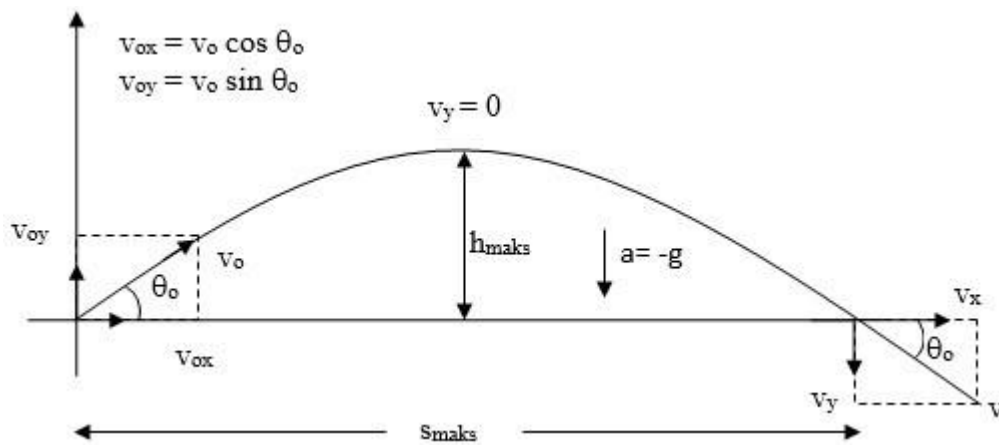


Figure 1. Trajectory of a projectile with an initial velocity v_0 and an elevation angle θ_0

As shown in Figure 1, the projectile's initial velocity (v_0) can be resolved into horizontal (v_{0x}) and vertical (v_{0y}) components. The horizontal velocity component remains constant because no force acts in the horizontal direction, whereas the vertical velocity component changes due to gravitational acceleration. According to Newton's Second Law, the force components acting on the projectile are expressed as follows,

$$F_x = 0 \text{ and } F_y = ma_y \quad (1)$$

therefore,

$$a_x = 0 \text{ and } a_y = -g \quad (2)$$

Where F_x is the net force in the horizontal direction (N), F_y is the net force in the vertical direction (N), a_x is the horizontal acceleration (m/s^2), a_y is the vertical acceleration (m/s^2) dan g is the acceleration due to gravity ($9,8 \text{ m/s}^2$).

The projectile's initial velocity (v_0) can be resolved into horizontal and vertical components. The horizontal component remains constant because it experiences no acceleration, whereas the vertical component changes with time under the effect of gravity. According to Serway and Jewett [2], these components can be expressed as follows.

$$v_x = v_0 \cos \theta_0 \quad (3)$$

$$v_y = v_0 \sin \theta_0 - gt \quad (4)$$

Where v_x is the horizontal velocity component (m/s), v_y is the vertical velocity component (m/s), v_0 is the projectile's initial velocity (m/s), θ_0 is the initial elevation angle ($^\circ$) dan t is the time of flight (s).

The projectile's position at any instant is described by its horizontal and vertical coordinates. Since the motion in the horizontal direction is uniform, whereas the motion in the vertical direction is uniformly accelerated, the projectile's position can be expressed as follows.

$$x = (v_0 \cos \theta_0)t \quad (5)$$

$$y = (v_0 \sin \theta_0)t - \frac{1}{2}gt^2 \quad (6)$$

In this experiment, Equation (5) is used to determine the projectile's initial velocity (v_0) based on the measured horizontal range and time of flight. In addition, the projectile's maximum height is determined using the following relationship:

$$h_{\text{maks}} = \frac{v_0^2 \sin^2 \theta_0}{2g} \quad (7)$$

Where h_{maks} adalah is the projectile's maximum height (m), v_0 adalah is the projectile's initial velocity (m/s), θ_0 is the initial elevation angle ($^\circ$) dan g is the acceleration due to gravity ($9,8 \text{ m/s}^2$).

METHODOLOGY

A. Tools and Materials

Listed below are the tools and materials used in the experiment:

Table 1. Tools and materials for the projectile motion experiment

Apparatus	Quantity	Function
Projectile launcher	1 set	Used to launch a projectile at a specified initial velocity and elevation angle.
Projectiles	2 pieces	Used as the test object launched by the projectile launcher.
Silica sand	1 set	Used as the landing surface for the projectile.
Tape measure	1 piece	Used to measure the projectile's horizontal range from the launch point to the point of impact.
Stopwatch	1 piece	Used to measure the projectile's time of flight from the launch point to the point of impact.
Lab jack	1 piece	Used to support and position the silica sand container during the experiment.

B. Procedure

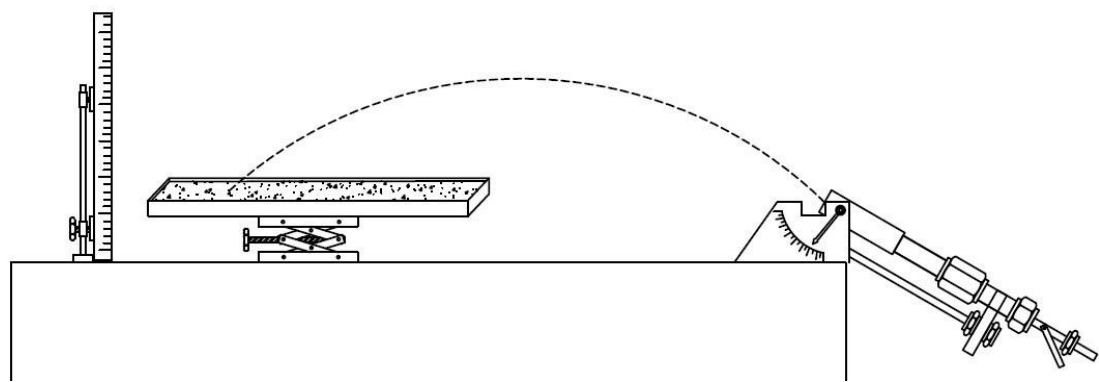


Figure 2. Setup of the projectile launcher equipment

The steps for this experiment are as follows.

1. Prepare the projectile launcher equipment.
2. Assemble the equipment as shown in Figure 2.
3. Set the elevation angle (θ_0) and secure it with the provided angle stop, according to the assistant’s briefing.
4. Adjust the height of the silica sand landing surface in the container to be level with the launcher’s height.
5. Insert the projectile load into the launcher.
6. Fire the projectile by pulling the trigger.
7. Use a stopwatch to determine the time (t) from when the projectile is fired until it hits the silica sand landing surface.
8. Use a tape measure to determine the horizontal distance (s) traveled by the projectile.
9. Repeat the experiment above 5 times.
10. Perform the experiment above with a projectile of a different mass.
11. Perform the experiment above with a different elevation angle (θ_0).

C. Experiment Flowchart

DATA ANALYSIS AND DISCUSSIONS

A. Data Analysis

The following are the tables of measurement results obtained in the laboratory:

Table 2. Data results for the small projectile at elevation angles of 30°, 45°, and 60°

Repetition	30° Elevation Angle Variation		45° Elevation Angle Variation		60° Elevation Angle Variation	
	$X_{maks} \text{ (m)}$	$t_{total} \text{ (s)}$	$X_{maks} \text{ (m)}$	$t_{total} \text{ (s)}$	$X_{maks} \text{ (m)}$	$t_{total} \text{ (s)}$
1						
2						
3						
4						
5						

Table 3. Data results for the large projectile at elevation angles of 30°, 45°, and 60°

Repetition	30° Elevation Angle Variation		45° Elevation Angle Variation		60° Elevation Angle Variation	
	$X_{maks} \text{ (m)}$	$t_{total} \text{ (s)}$	$X_{maks} \text{ (m)}$	$t_{total} \text{ (s)}$	$X_{maks} \text{ (m)}$	$t_{total} \text{ (s)}$
1						
2						
3						
4						
5						

Approved by:

Laboratory Assistant

B. Calculation Analysis

(Calculate the initial velocity of the projectile (v_0) and its maximum height ($h_{\max}$) using Equations 5 and 7!)

The following is a sample calculation based on experimental data.

Sample Calculation:

The final calculation results for all data obtained are as follows.

Table 4. Calculations results for the small projectile at elevation angles of 30°, 45°, and 60°

Repetition	30° Elevation Angle Variation		45° Elevation Angle Variation		60° Elevation Angle Variation	
	v_0 (m/s)	h_{maks} (m)	v_0 (m/s)	h_{maks} (m)	v_0 (m/s)	h_{maks} (m)
1						
2						
3						
4						
5						
Average						
Standard Deviation						
Uncertainty						

Table 5. Calculations results for the large projectile at elevation angles of 30°, 45°, and 60°

Repetition	30° Elevation Angle Variation		45° Elevation Angle Variation		60° Elevation Angle Variation	
	v_0 (m/s)	h_{maks} (m)	v_0 (m/s)	h_{maks} (m)	v_0 (m/s)	h_{maks} (m)
1						
2						
3						
4						
5						
Average						
Standard Deviation						
Uncertainty						

C. Graphs

(Plot the parabolic trajectories of the projectile at elevation angles of 30°, 45°, and 60°!)

D. Discussions

CONCLUSIONS

From this experiment, it can be concluded that:

1.

2.

3.

REFERENCES

- [1] D. Halliday, R. Resnick, and J. Walker, Fundamentals of Physics, 9th ed. Hoboken, NJ, USA: John Wiley & Sons, 2010.
- [2] R. A. Serway and J. W. Jewett, Physics for Scientists and Engineers with Modern Physics, 9th ed. Stamford, CT, USA: Cengage Learning, 2014.

PRELIMINARY ASSIGNMENT

1. Show that the trajectory of projectile motion is a parabola and provide examples of parabolic motion phenomena in daily life.
2. Show the equations for the position and velocity components in projectile motion.
3. Does the mass of the projectile affect its range, height, and velocity? Explain!
4. Do the launch angle (θ_0), initial velocity (v_0), and initial height (y_0) affect the range and time of flight of the projectile? Explain!